

Robust Tensor Learning with Graph Diffusion for Scalable Multi-view Graph Clustering

Anonymous Author(s)

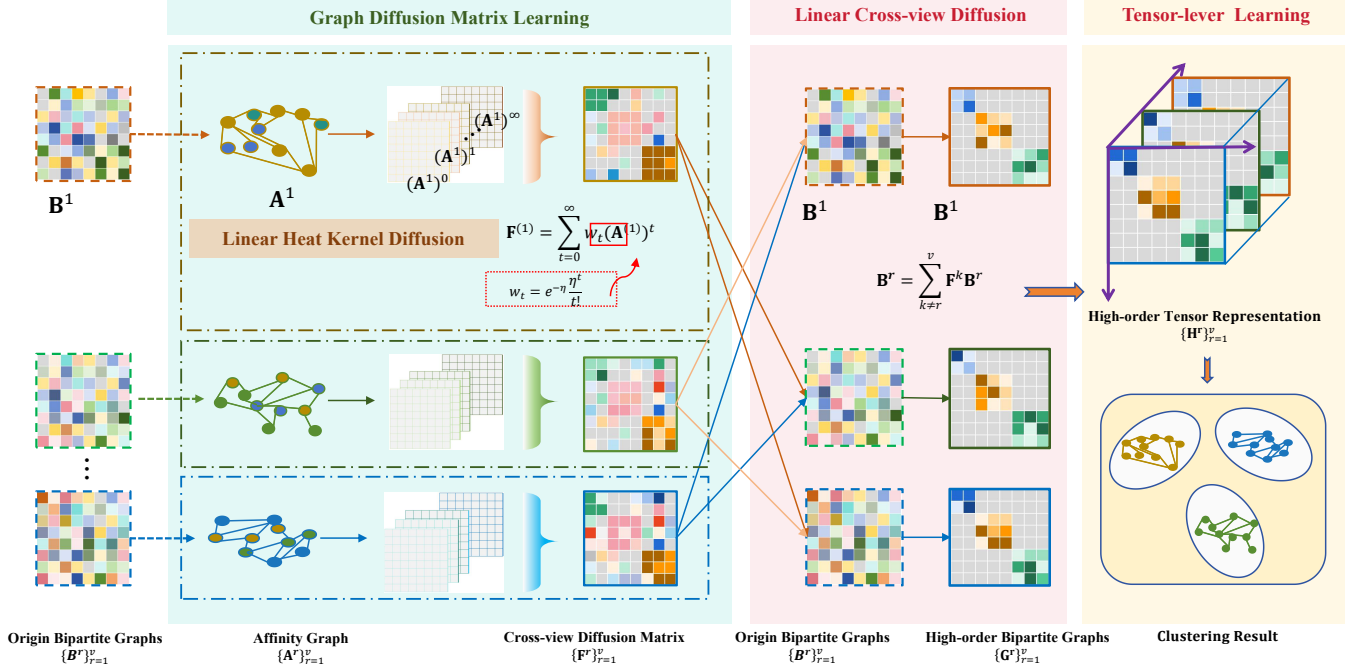


Figure 1: The framework of RTGD-MVC. By introducing a graph diffusion mechanism, RTGD-MVC effectively suppress the propagation and accumulation of graph structure noise during the learning process. Through linear transformation operations, we optimize the traditional spectral diffusion method, reducing its complexity from $O(n^3)$ to $O(n)$. This not only improves the performance of multi-view clustering but also significantly enhances computational efficiency. The code is publicly available at <https://anonymous.4open.science/r/RTGD-MVC-6646/>.

Abstract

The rapid proliferation of multi-view data has necessitated robust and scalable clustering techniques capable of capturing complex, high-dimensional patterns. While Multi-view Bipartite Graph Clustering (MVBGC) has shown promising results, existing approaches often overlook that the generated bipartite graph is susceptible to disturbances from complex structures and noise. To address these challenges, we propose RTGD-MVC, a novel framework for Robust Tensor Learning with Graph Diffusion tailored for efficient and scalable multi-view graph clustering. RTGD-MVC integrates a graph diffusion mechanism to suppress noise propagation and employs cross-view diffusion to enhance global consistency while

capturing complementary information across views. Additionally, a non-convex Tensor Exponential Norm (TEN) is introduced as a tighter surrogate for the tensor rank, enabling the learning of more discriminative and noise-robust representations. By embedding these components into a unified optimization model with linear computational complexity, RTGD-MVC achieves both theoretical efficiency and practical scalability. Extensive experiments on diverse benchmark datasets demonstrate that RTGD-MVC significantly outperforms state-of-the-art methods, highlighting its superior ability to capture intricate multi-view correlations and structural patterns.

CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper;** *Generate the Correct Terms for Your Paper;* Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

Keywords

Multimodal Fusion, Tensor Learning, Noise Robustness, Multi-view Clustering

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

Conference acronym 'XX, Woodstock, NY

© 2018 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 978-1-4503-XXXX-X/2018/06

<https://doi.org/XXXXXX.XXXXXXX>

ACM Reference Format:

Anonymous Author(s). 2018. Robust Tensor Learning with Graph Diffusion for Scalable Multi-view Graph Clustering. In *Proceedings of Make sure to enter the correct conference title from your rights confirmation email (Conference acronym 'XX)*. ACM, New York, NY, USA, 9 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

With the widespread adoption of the internet, social media, and the Internet of Things, data from multiple sources is rapidly increasing. Multi-view Clustering (MVC) can help uncover underlying patterns and structures in data from diverse sources, revealing hidden relationships. Given the flexibility and powerful ability of graphs to represent complex data structures, Multi-view Graph Clustering (MVG) is emerging as a rapidly expanding field within MVC. Multi-view Bipartite Graph Clustering (MVBGC) is developed by establishing correlations between $m(m \ll n)$ anchors and n instances, forming a $n \times m$ bipartite graph [38, 40, 41]. This approach reduces complexity to $O(mn)$, significantly enhancing scalability.

The success of MVBGC heavily depends on the quality of the bipartite graph [15]. Many MVBGC models have been proposed by various strategies to construct a “nice” bipartite graph, such as using sampling [9, 42] or optimizing manners [14, 16, 41] to select representative anchors, introducing different regularizations [2] or constraints [17] to refine intrinsic structures, or concatenating multi-scale bipartite graphs [12, 43, 44] across multiple views to achieve ensemble clustering. While clustering using the generated bipartite graph is efficient, it comes with several inherent limitations. Primarily, directly generated bipartite graphs capture only a limited amount of local information from the data, making them susceptible to disturbances from complex structures and noise. Furthermore, they often lack global consistency, leading to sensitivity in clustering outcomes, particularly in cases of non-uniform data distribution. The bipartite graph is generated independently row by row. However, the global interconnections among these rows are overlooked. To address this limitation, we introduce high-order connections by employing the heat-kernel diffusion procedure to bridge this gap. This procedure characterizes immediate information by facilitating message passing between neighboring anchors.

Although MVBGC methods have achieved promising performance, there are still significant limitations for improvement. Firstly, existing MVGC methods focus solely on exploiting dyadic relationships, such as the pairwise correlation between individual samples [4, 31, 41] and the pairwise correlation between distinct views [5, 23, 39], ignoring the valuable higher-dimensional correlations that exist among multiple samples and views. By failing to exploit these high-dimensional dependencies, existing techniques miss out on vital information that could significantly enhance clustering performance and uncover more intricate patterns within the data. Secondly, most MVBGC methods typically constructed bipartite graphs for each view [45] and only the first-order similarity information between samples, which ignores high-order information, hindering clustering performance.

Low-rank tensors are effective in capturing intrinsic structures and uncovering high-dimensional correlations among multi-view representations [1, 13, 36]. Existing methods commonly adopt the

tensor nuclear norm (TNN) as a convex surrogate for the non-convex tensor rank function [6, 8]. However, a notable limitation of TNN lies in its equal treatment of all singular values, which can lead to an over-penalization of the dominant rank components [10, 30]. This uniform penalization scheme may result in a sub-optimal tensor representation in each view.

To tackle the aforementioned challenges and limitations, we propose a novel and efficient multi-view graph clustering framework termed Robust Tensor Learning with Graph Diffusion for Scalable Multi-view Graph Clustering (RTGD-MVC). This innovative approach seamlessly integrates graph diffusion, tensor learning, consistency learning, and complementarity learning into a unified optimization model, enabling a more comprehensive and accurate capture of the intricate patterns within multi-view data. Specifically, we employ the within-view graph diffusion to derive the high-order bipartite graphs. Then, enhanced consistency is derived by the cross-view diffusion, and complementarity is captured across views. By propagating information along the edges of the bipartite graph, the diffusion process strengthens the connections between samples and their respective representative anchors. This leads to more compact and cohesive data representations, where samples with similar patterns are brought closer together, effectively capturing the intrinsic geometry of the data manifold within each view. To effectively utilize high-dimensional correlation among multi-view representations, we present a Tensor Exponential Norm (TEN) for MVBGC. Non-convex TEN is a tighter surrogate of tensor rank, which is superior to the convex TNN. Hence, more discriminative representations can be learned for MVBGC. The novelty and contributions of this paper are summarized as follows.

- By incorporating a graph diffusion mechanism, RTGD-MVC effectively mitigates the propagation and accumulation of graph structure noise during the learning process while capturing higher-order connections. Specifically, we apply within-view graph diffusion to generate high-order bipartite graphs. This is followed by cross-view diffusion to enhance consistency and capture complementarity across views. Through the introduction of linear transformation operations, we optimize the traditional spectral diffusion method, reducing its complexity from $O(n^3)$ to $O(n)$.
- By constructing a tensor-based representation from high-order bipartite graphs via graph diffusion, RTGD-MVC effectively encodes complex interactions among multiple samples and views, uncovering high-dimensional correlations and capturing the rich patterns. The non-convex TEN serves as a tighter surrogate for the tensor rank, offering advantages over the convex TNN. Hence, more discriminative representations can be learned for MVBGC.
- Integration of MVBGC, graph-level high-order graph diffusion, tensor-level high-dimensional improved tensor constraints into a unified framework for comprehensive multi-view data pattern capture. We proposed an efficient algorithm designed to address our optimization problem, characterized by its low computational complexity and rigorously proven convergence. Comprehensive experimental evaluations on nine challenging datasets further underscore the superiority of our model.

2 RELATED WORK

2.1 Multi-view Bipartite Graph Clustering

Recent advances in multi-view graph clustering research have introduced anchor-based bipartite graph to overcome the computational challenges inherent in traditional methods when dealing with large-scale datasets [12, 26, 34, 43]. The fundamental principle of Multi-view Bipartite Graph Clustering (MVBGC) involves the strategic selection of a representative subset of anchors from the complete sample collection, with the anchor count being substantially smaller than the total sample population [10, 23, 25, 28]. This approach represents a significant departure from conventional techniques that necessitate the computation of relationships between all sample pairs. Instead, anchor-based methods restrict similarity calculations to anchor-sample pairs exclusively, thereby substantially reducing the computational complexity associated with relationship graph construction. Furthermore, the implementation of dimensionally-reduced graphs contributes to additional computational efficiency, enabling more effective multi-view integration and clustering operations while maintaining lower processing requirements. This paradigm shift in approach has proven particularly valuable for applications involving high-dimensional or voluminous datasets [18, 19]. The success of MVBGC depends on the quality of the bipartite graph. Although various strategies have been proposed to construct effective graphs, these methods still face limitations. Directly generated bipartite graphs capture limited local information, making them sensitive to noise and lacking global consistency. To address this, we propose a graph diffusion mechanism that facilitates message passing between neighboring anchors. This approach effectively mitigates the propagation and accumulation of noise in the graph structure during the learning process, thereby yielding more robust global structure representations.

2.2 Low-rank Tensor Learning

The low-rank tensor can effectively capture the underlying structure and uncover higher-order correlations among multi-view representations, making it widely used in multi-view clustering [7, 8, 10, 11]. Prevalently used tensor rank approximations (e.g., TNN) typically under-penalize small singular values while over-penalizing large ones, leading to information loss and residual noise. To address this, rather than relying on TNN to approximate the tensor rank, we propose a noise-robust enhanced Tensor Exponential Norm (TEN). Specifically, the proposed TEN not only preserves the larger singular values, which encode valuable structural information, but also eliminates the smaller ones that contain redundant information. The non-convex TEN provides a more accurate surrogate for the tensor rank, offering distinct advantages over the convex TNN. This enables the learning of more discriminative representations for MVBGC.

A comparative analysis with established tensor rank approximation methods—TNN [37], ETR [10], TLSpN [8], and GER [6] demonstrates the superiority of TEN (Fig. 2). TEN more accurately approximates the true rank, especially for near-zero singular values. As a result, minimizing TEN leads to a noise-robust model that effectively eliminates noise while preserving high-order consistency across views.

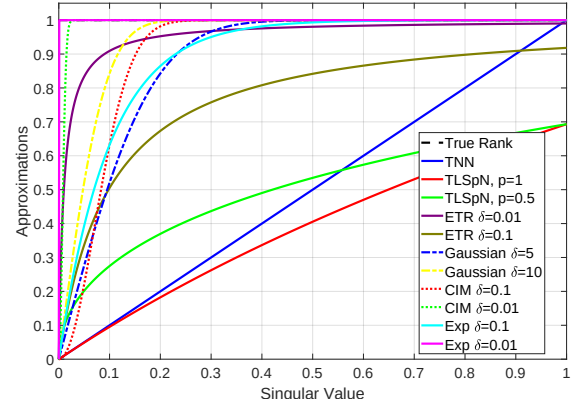


Figure 2: Comparison of different methods to approximate the true rank function. The true rank function is represented by a horizontal line at a value of 1 on the vertical axis

3 PROPOSED METHOD

3.1 Preliminaries

In this paper, bold lowercase $\mathbf{x}$ represents vectors, while lowercase letters x denote scalars. Bold uppercase letters $\mathbf{X}$ indicate the matrix. $\mathbf{I}_n$ represents the $n \times n$ identity matrix. Calligraphy letters denote tensors, such as $\mathcal{X} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$. $\mathcal{X}^k$ refers to the k -th frontal slice of $\mathcal{X}$ and $\mathcal{F}$ represents the Fast Fourier Transformation (FFT) along the third dimension of tensor $\mathcal{X}$. A multi-view dataset with n samples and v views is denoted as $\{\mathbf{X}^r\}_{r=1}^v$, where $\mathbf{X}^r \in \mathbb{R}^{n \times d^r}$, d^r is the feature dimension of the r -th view. c represents the number of clusters, and m denotes the number of anchors in the dataset.

3.2 Linear Cross-view Graph Diffusion

In this subsection, we proposed a novel linear graph diffusion approach in cross-view to learn a more discriminative and diverse graph-level high-order bipartite graph. RTGD-MVC is the first to introduce a linear method for heat kernel diffusion, significantly reducing the time complexity from $O(n^3)$ to $O(n)$. This makes our approach highly scalable and efficient.

To achieve this, we apply the k -means clustering algorithm to each view $\mathbf{X}^r$ of the original dataset, generating m ($m \ll n$) anchors $\mathbf{C}^r = [\mathbf{c}_1^r, \mathbf{c}_2^r, \dots, \mathbf{c}_m^r] \in \mathbb{R}^{m \times d^r}$, uniformly. Next we construct a bipartite graph $\mathbf{B}^r \in \mathbb{R}^{n \times m}$ between the anchors and the original data points to capture the geometry structure of the r -th view. Specifically, each row of the bipartite graph $\mathbf{B}^r$ is learned by solving the following optimization problem:

$$\min_{\mathbf{s}_i^r, \mathbf{x}_i^r \geq 0} \sum_{j=1}^m \|\mathbf{x}_i^r - \mathbf{c}_j^r\|^2 s_{ij}^r + \gamma \sum_{j=1}^m (s_{ij}^r)^2, \quad (1)$$

where $\mathbf{s}_i^r$, $\mathbf{x}_i^r$ and $\mathbf{a}_j^r$ represent the i -th row of $\mathbf{S}^r$, $\mathbf{X}^r$ and $\mathbf{C}^r$, respectively. γ is the trade-off parameter.

To avoid the need to search for γ , we constrain each row to have only k non-zero elements, which is set to $k = 10$ in this paper. The

closed-form solution is given by[29]:

$$s_{ij}^r = \begin{cases} \frac{||\mathbf{x}_i^r - \mathbf{c}_{(k+1)}^r||^2 - ||\mathbf{x}_i^r - \mathbf{c}_j^r||^2}{k||\mathbf{x}_i^r - \mathbf{c}_{(k+1)}^r||^2 - \sum_{k'=1}^k ||\mathbf{x}_i^r - \mathbf{c}_{k'}^r||^2}, & j \leq k, \\ 0, & j > k. \end{cases} \quad (2)$$

The bipartite graph $\mathbf{B}^r$ primarily captures the local structure of data, we proposed a linear heat kernel diffusion approach to enhance the graph by incorporating high-order and global inter-connections. This is achieved through a weighted combination of all items. The general form of heat kernel diffusion [3] can be expressed as:

$$\mathbf{F}^r = \sum_{t=0}^{\infty} \gamma_t \left(\mathbf{P}^r \mathbf{P}^{rT} \right)^t, \quad (3)$$

where $\mathbf{P}^r = \mathbf{B}^r \mathbf{D}^{r-\frac{1}{2}}$, $\mathbf{D}^r$ is the column degree matrix of $\mathbf{B}^r$, defined as $\mathbf{D}^r = \text{diag}(\mathbf{B}^r \mathbf{1})$. $\mathbf{A}^r = \mathbf{P}^r \mathbf{P}^{rT}$, is the affinity matrix. The weights for the heat kernel diffusion are computed as

$$\gamma_t = e^{-\eta} \frac{\eta^t}{t!}, \quad (4)$$

where η is a non-negative parameter that controls the decay speed.

By combining Eq. (3) with the weighting scheme in Eq. (4), the general heat kernel diffusion matrix $\mathbf{F}^r$ from the r -th view is instantiated as follows:

$$\mathbf{F}^r = e^{-\eta} \sum_{t=0}^{\infty} \frac{\eta^t}{t!} \left(\mathbf{P}^r \mathbf{P}^{rT} \right)^t = \exp(-\eta \mathbf{L}^r), \quad (5)$$

where $\mathbf{L}^r$ is the Laplacian matrix associated with the r -th bipartite graph, defined as $\mathbf{L}^r = \mathbf{I}_n - \mathbf{P}^r \mathbf{P}^{rT}$. We utilize cross-view graph diffusion to derive high-order bipartite graphs, effectively capturing the intrinsic geometry in a more discriminative and diverse manner. The cross-view diffusion process can be described as follows:

$$\tilde{\mathbf{B}}^r = \frac{1}{v-1} \sum_{k \neq r}^v \mathbf{F}^k \mathbf{B}^r. \quad (6)$$

Through cross-view graph diffusion, the resulting enhanced high-order bipartite graphs become more discriminative and diverse. We are the first to propose a linear formulation for optimizing the heat kernel diffusion, which can be obtained by the following theorem:

THEOREM 1. Suppose the graph diffusion matrix $\mathbf{F} \in \mathbb{R}^{n \times n}$ is derived from the bipartite graph $\mathbf{B} \in \mathbb{R}^{n \times m}$. The graph diffusion matrix can be expressed as follows:

$$\mathbf{F} = e^{-\eta} \mathbf{I}_n - \mathbf{P} \hat{\mathbf{F}} \left(\mathbf{P}^T \mathbf{P} \right)^{-1} \mathbf{P}^T, \quad (7)$$

where $\mathbf{P} = \mathbf{B} \mathbf{D}^{-\frac{1}{2}}$, $\mathbf{D}$ is the column degree matrix of $\mathbf{B}$, defined as $\mathbf{D} = \text{diag}(\mathbf{B} \mathbf{1})$, $\hat{\mathbf{F}} = e^{-\eta} \mathbf{I}_m - \exp(-\eta \mathbf{L}_m)$, $\mathbf{L}_m = \mathbf{I}_m - \mathbf{P}^T \mathbf{P}$.

The detailed proof of Theorem 1 is provided in the Supplementary. According to Theorem 1, the optimization of the heat kernel cross-view diffusion could be demonstrated as follows:

$$\tilde{\mathbf{B}}^r = \sum_{k \neq r}^v \left(e^{-\eta} \mathbf{I}_n - \mathbf{P}^k \hat{\mathbf{F}}^k \left(\mathbf{P}^{kT} \mathbf{P}^k \right)^{-1} \mathbf{P}^{kT} \right) \mathbf{B}^r. \quad (8)$$

We can achieve linear cross-view graph diffusion through the following transformation operation:

$$\tilde{\mathbf{B}}^r = \sum_{k \neq r}^v \left(e^{-\eta} \mathbf{B}^r - \mathbf{P}^k \hat{\mathbf{F}}^k \left(\mathbf{P}^{kT} \mathbf{P}^k \right)^{-1} \left(\mathbf{P}^{kT} \mathbf{B}^r \right) \right). \quad (9)$$

Through the above linear cross-view graph diffusion, we can learn the more discriminative and diverse graph-level high-order bipartite graphs $\{\tilde{\mathbf{B}}^r\}_{r=1}^v$.

3.3 Tensor-level High-dimensional Representation Learning

Tensors have a superior ability to explore high-order consistent information among views. We learn a tensor-level high-dimensional representation $\mathcal{H}$ based on the enhanced graph-level high-order bipartite graph obtained by linear cross-view diffusion as follows:

$$\begin{aligned} \min_{\{\mathbf{H}^r, \mathbf{E}^r\}} \quad & \sum_{r=1}^v \|\mathbf{E}^r\|_{2,1} \\ \text{s.t. } \forall r, \quad & \mathbf{H}^r = \tilde{\mathbf{B}}^r - \mathbf{E}^r, \\ & \mathcal{H} = \Phi(\mathbf{H}^1, \dots, \mathbf{H}^v), \end{aligned} \quad (10)$$

where $\mathbf{E}^r$ represents the view-specific noises. As defined in [37], $\Phi(\cdot)$ merges view-specific representations $\mathbf{H}^r$ into an $m \times v \times n$ tensor $\mathcal{H}$ and its inverse function is $\Phi_r^{-1}(\mathcal{H}) = \mathbf{H}^r$.

Moreover, we impose a low-rank constraint on tensor $\mathcal{H}$ to learn the tensor-level high-dimensional consistency among views. Prevalently adopted tensor rank approximations (e.g., TNN) usually under-penalize small singular values and over-penalize large singular values, resulting in information loss and noise residuals. Instead of adopting the TNN to approximate the tensor rank, we design a noise-robust enhanced Tensor Exponential Norm (TEN) which is defined in Definition 1.

DEFINITION 1. Given a tensor $\mathcal{H} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$, the Tensor Exponential Norm is defined as:

$$\|\mathcal{H}\|_{TEN} = \frac{1}{n_3} \sum_{k=1}^{n_3} \|\mathcal{H}_f^k\|_{TEN} = \frac{1}{n_3} \sum_{k=1}^{n_3} \sum_{i=1}^h \left(1 - e^{-\frac{S_f^k(i,i)}{\delta}} \right), \quad (11)$$

where $0 \leq \delta \leq 1$, $h = \min(n_1, n_2)$ and S_f is obtained by the t -SVD of $\mathcal{H}_f = \mathcal{U}_f S_f \mathcal{V}_f^T$ in Fourier domain. $\mathcal{H}_f^k$ denotes the k -th frontal slice of the tensor $\mathcal{H}_f$.

REMARK 1. Superiority of TEN: Inspired by the Exponential function [27], we proposed a novel approximation function $f_{TEN}(x) = 1 - e^{-\frac{x}{\delta}}$, $0 \leq \delta \leq 1$. Notably $f_{TEN}(0) = 0$, aligning with the true rank function. Comparative analysis with established tensor rank approximation methods-TNN [37], ETR[10], TLS_pN[8] and GER[6] demonstrates TEN's superiority (Fig. 2). TEN approximates the true rank more accurately, particularly for near-zero singular values. Consequently, minimizing $\|\mathcal{H}\|_{TEN}$ yields a noise-robust model, eliminating noise while preserving high-order consistency among views.

With the inclusion of cross-view diffusion and the novel TEN-based low-rank regularization, the RTGD-MVC model learns a high-order bipartite graph representation across views. This model incorporates enhanced high-order bipartite graphs $\{\tilde{\mathbf{B}}^r\}_{r=1}^v$ from linear

Table 1: Multi-view Clustering Results Comparison (Mean $\pm$ Standard Deviation, reported as percentages %)

Data set	Metric(%)	OPMC	FPMVS-CAG	FMVACC	FDAGF	AWMVC	FastMICE	TLsPNM	S2MVTC	RCAGL	Ours
BBC	ACC	54.53(6.95)	32.26(0)	80.63(6.35)	88.28(2.05)	64.93(0.29)	41.45(5.51)	99.42(0)	-	53.42(7.02)	100(0)
	NMI	42.65(5.46)	2.98(0)	60.93(4.23)	73.85(2.64)	40.84(0.46)	22.43(7.09)	98.07(0)	-	36.72(5.85)	100(0)
	PUR	60.88(5.46)	37.37(0)	81.72(4.23)	57.36(2.64)	64.93(0.46)	49.94(7.09)	99.42(0)	-	63.28(5.85)	100(0)
BDGP	ACC	57.84(6.06)	55.60(0)	67.52(4.13)	95.28(0.29)	93.64(0)	79.98(7.49)	97.00(0)	79.92(0)	51.78(0.02)	100(0)
	NMI	48.43(3.23)	37.56(0)	49.56(5.84)	88.60(0.49)	82.87(0)	65.66(6.53)	90.16(0)	89.76(0)	34.49(0.04)	100(0)
	PUR	59.60(4.19)	58.96(0)	67.98(3.95)	95.28(0.29)	93.64(0)	80.36(7.01)	97.00(0)	79.92(0)	52.58(0.02)	100(0)
CCV	ACC	15.93(0.36)	23.00(0)	19.93(0.87)	19.99(0)	18.89(0.24)	20.22(0.99)	59.84(0)	56.02(0)	23.47(0.59)	95.91(0)
	NMI	13.18(0.53)	16.98(0)	16.08(0.57)	16.87(0.04)	13.98(0.14)	15.68(1.20)	52.80(0)	61.47(0)	18.91(0.25)	95.09(0)
	PUR	20.48(0.43)	25.07(0)	23.38(0.74)	23.53(0.04)	22.04(0.23)	24.13(0.94)	61.32(0)	58.31(0)	26.65(0.54)	96.57(0)
Caltech101	ACC	51.57(4.46)	57.12(0)	40.64(1.47)	68.98(8.74)	43.94(1.36)	54.38(1.17)	62.82(0)	67.44(0)	84.78(8.45)	79.04(0)
	NMI	38.42(1.09)	55.57(0)	40.64(3.84)	65.16(2.98)	43.26(0.66)	61.65(1.95)	72.84(0)	73.96(0)	84.78(7.53)	83.45(0)
	PUR	80.72(0.31)	86.57(0)	81.46(1.44)	86.75(2.20)	81.89(0.30)	86.85(0.84)	92.61(0)	91.32(0)	89.35(2.56)	96.74(0)
small-NUS	ACC	15.25(0.54)	20.70(0)	13.88(0.43)	15.57(0.34)	16.57(0.25)	16.52(1.18)	28.60(0)	69.05(0)	18.38(0.78)	63.25(0)
	NMI	20.16(0.39)	19.23(0)	16.95(0.55)	18.54(0.29)	19.68(0.19)	19.87(0.88)	29.66(0)	81.56(0)	19.42(0.37)	92.63(0)
	PUR	25.88(0.46)	24.10(0)	23.48(0.68)	24.84(0.41)	26.04(0.20)	25.42(1.18)	36.80(0)	74.10(0)	25.55(0.43)	85.86(0)
scene-15	ACC	29.68(0.68)	33.47(0)	34.32(4.46)	30.70(0.10)	37.73(0.80)	42.85(0.71)	93.82(0)	79.18(0)	36.96(2.12)	97.21(0)
	NMI	29.53(0.53)	35.45(0)	31.76(3.81)	28.37(0.22)	34.62(0.56)	41.21(0.86)	89.54(0)	86.13(0)	38.62(1.21)	93.54(0)
	PUR	32.95(0.83)	35.07(0)	37.09(4.88)	32.35(0.09)	41.55(1.06)	46.03(0.73)	93.82(0)	81.94(0)	39.12(2.14)	95.83(0)
Reuters	ACC	43.75(1.11)	57.52(0)	58.83(2.37)	61.19(0.43)	OM	36.48(0.54)	OM	85.40(0)	62.37(3.27)	99.84(0)
	NMI	28.64(1.64)	35.96(0)	41.34(1.87)	42.92(0.32)	OM	26.16(0.27)	OM	82.40(0)	43.61(2.36)	99.48(0)
	PUR	53.77(1.22)	62.34(0)	65.75(2.54)	68.98(0.57)	OM	47.95(0.62)	OM	85.40(0)	66.67(3.35)	99.84(0)
Nuswideobj	ACC	14.16(0.20)	19.22(0)	12.26(0.21)	13.38(0.17)	14.61(0.13)	15.74(0.16)	OM	67.58(0)	19.36(0.49)	89.32(0)
	NMI	14.70(0.22)	13.52(0)	11.13(0.18)	11.96(0.17)	12.80(0.13)	15.40(0.15)	OM	78.58(0)	13.71(0.23)	91.54(0)
	PUR	26.12(0.26)	23.67(0)	22.30(0.24)	23.99(0.02)	24.32(0.06)	26.11(0.17)	OM	73.73(0)	24.86(0.51)	87.54(0)
WebKB	ACC	93.22(8.82)	94.96(0)	93.35(1.09)	94.75(0.15)	79.16(0)	91.54(11.48)	96.29(0)	99.71(0)	85.49(0.44)	100(0)
	NMI	63.01(22.02)	69.99(0)	56.11(4.80)	63.69(1.09)	11.58(0)	59.88(18.11)	76.74(0)	96.76(0)	32.45(1.44)	100(0)
	PUR	94.22(5.66)	94.96(0)	93.35(1.09)	94.75(0.15)	79.16(0)	93.46(5.42)	96.29(0)	99.71(0)	91.87(0.43)	100(0)
small-Reuters	ACC	41.72(3.77)	38.60(0)	57.37(2.60)	50.40(3.73)	43.29(0.83)	41.65(2.75)	75.13(0)	81.47(0)	45.57(2.06)	100(0)
	NMI	26.39(2.45)	13.25(0)	33.50(0.64)	34.56(4.57)	25.95(0.82)	26.38(1.80)	74.14(0)	76.43(0)	23.39(1.02)	100(0)
	PUR	51.68(1.15)	46.93(0)	62.09(1.08)	49.13(3.24)	51.84(0.40)	50.37(1.85)	82.87(0)	81.47(0)	52.06(1.04)	100(0)
YoutubeFace	ACC	23.40(0.26)	24.10(0)	24.54(1.57)	14.58(0.42)	OM	22.76(2.06)	OM	55.24(0)	30.31(0.46)	88.47(0)
	NMI	23.15(0.17)	24.40(0)	22.11(1.24)	12.29(0.35)	OM	20.31(1.90)	OM	76.07(0)	26.29(0.45)	91.09(0)
	PUR	33.14(0.21)	32.74(0)	34.07(1.43)	28.40(0.14)	OM	33.16(1.46)	OM	74.93(0)	33.97(0.25)	89.68(0)

Note: Each method is independently run 10 times. **Red bold** indicates the highest value, and **blue bold** indicates the second-highest. Missing values ('-' or 'OM') are due to out-of-memory errors or unavailable results. For methods like FPMVS-CAG and Ours, zero standard deviation implies deterministic outputs.

cross-view graph diffusion, yielding a denoised, view-specific representation $\mathbf{H}^r$ that integrates complementary information across views. The final tensor-level representation $\mathcal{H}$ captures complex interactions and high-dimensional correlations among samples and views. The RTGD-MVC model is formulated as follows, with its comprehensive framework depicted in Fig. 1:

$$\begin{aligned}
& \min_{\{\mathbf{H}^r, \mathbf{E}^r\}} \|\mathcal{H}\|_{\text{TEN}} + \lambda \sum_{r=1}^v \|\mathbf{E}^r\|_{2,1} \\
& \text{s.t. } \forall r, \quad \mathbf{H}^r = \tilde{\mathbf{B}}^r - \mathbf{E}^r, \\
& \quad \mathcal{H} = \Phi(\mathbf{H}^1, \dots, \mathbf{H}^v),
\end{aligned} \tag{12}$$

where $\lambda > 0$ is a regularization parameter that controls the trade-off between the tensor norm and the noise term.

4 Optimization

Leveraging the Alternating Direction Method of Multipliers[20] (ADMM), we develop an efficient optimization strategy for Eq. (12). By introducing an auxiliary variable $\mathcal{G}$, we reformulate the problem into an unconstrained Lagrangian form:

$$\begin{aligned}
& \mathcal{L}(\{\mathbf{H}^r\}_{r=1}^v, \{\mathbf{S}^r\}_{r=1}^v, \mathcal{G}, \mathcal{Y}) \\
& = \|\mathcal{G}\|_{\text{TEN}} + \lambda \sum_{r=1}^v \|\mathbf{E}^r\|_{2,1} \\
& + \sum_{r=1}^v \langle \mathbf{S}^r, \frac{1}{v-1} \sum_{k \neq r}^v (\mathbf{F}^k \mathbf{B}^r) - \mathbf{H}^r - \mathbf{E}^r \rangle \\
& + \frac{\mu}{2} \sum_{v=1}^r \left\| \frac{1}{v-1} \sum_{k \neq r}^v (\mathbf{F}^k \mathbf{B}^r) - \mathbf{H}^r - \mathbf{E}^r \right\|_{\text{F}}^2 \\
& + \langle \mathcal{Y}, \mathcal{H} - \mathcal{G} \rangle + \frac{\rho}{2} \|\mathcal{H} - \mathcal{G}\|_{\text{F}}^2,
\end{aligned} \tag{13}$$

where $\mathcal{Y}$ and $\{\mathbf{S}^r\}_{r=1}^v$ denote Lagrange multipliers, μ and ρ are positive penalty parameters governing convergence. The variables in Eq. (13) are then updated alternately.

Update $\mathbf{H}^r$ With other variables fixed, and using $\Phi_r^{-1}(\mathcal{Y}) = \mathbf{Y}^r$ and $\Phi_r^{-1}(\mathcal{G}) = \mathbf{G}^r$, we optimize $\mathbf{H}^r$ through the following sub-problem:

$$\begin{aligned}
& \min_{\mathbf{H}^r} \langle \mathbf{S}^r, \tilde{\mathbf{F}}^r \mathbf{B}^r - \mathbf{H}^r - \mathbf{E}^r \rangle + \langle \mathbf{Y}^r, \mathbf{H}^r - \mathbf{G}^r \rangle \\
& + \frac{\mu}{2} \|\tilde{\mathbf{F}}^r \mathbf{B}^r - \mathbf{H}^r - \mathbf{E}^r\|_{\text{F}}^2 + \frac{\rho}{2} \|\mathbf{H}^r - \mathbf{G}^r\|_{\text{F}}^2,
\end{aligned} \tag{14}$$

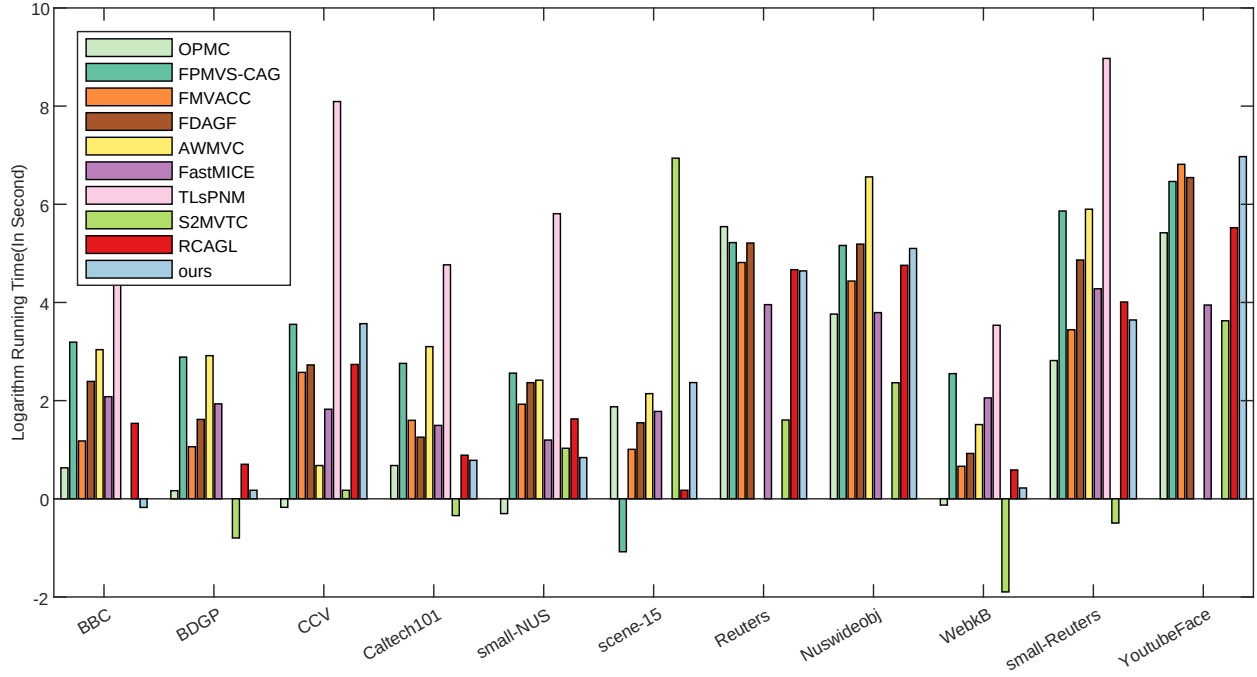


Figure 3: Logarithm of the running time on all data sets (in Sec.).

where $\tilde{\mathbf{F}}^r = \left(\frac{1}{v-1} \sum_{k \neq r}^v \mathbf{F}^k \right)$. This quadratic, smooth, and convex problem yields the closed-form solution:

$$\mathbf{H}^r = \frac{\mu \tilde{\mathbf{F}}^r \mathbf{B}^r - \mu \mathbf{E}^r + \rho \mathbf{G}^r + \mathbf{S}^r - \mathbf{Y}^r}{\mu + \rho} \quad (15)$$

Update $\mathbf{E}^r$: With other variables fixed, the sub-problem for $\mathbf{E}^r$ is:

$$\begin{aligned} \min_{\mathbf{E}^r} \quad & \langle \mathbf{Y}^r, \tilde{\mathbf{F}}^r \mathbf{B}^r - \mathbf{H}^r - \mathbf{E}^r \rangle + \lambda \|\mathbf{E}^r\|_{2,1} \\ & + \frac{\mu}{2} \|\tilde{\mathbf{F}}^r \mathbf{B}^r - \mathbf{H}^r - \mathbf{E}^r\|_F^2 \\ \Leftrightarrow \min_{\mathbf{E}^r} \quad & \frac{\lambda}{\mu} \|\mathbf{E}^r\|_{2,1} + \frac{1}{2} \|\mathbf{E}^r - \tilde{\mathbf{E}}^r\|_F^2, \end{aligned} \quad (16)$$

where $\tilde{\mathbf{E}}^r = \tilde{\mathbf{F}}^r \mathbf{B}^r - \mathbf{H}^r + \frac{1}{\mu} \mathbf{S}^r$. The solution is obtained via the $l_{2,1}$ minimization thresholding operator[21]:

$$[\mathbf{E}^r]_{:,j} = \frac{(\|[\tilde{\mathbf{E}}^r]_{:,j}\|_2 - \lambda/\mu)}{\|[\tilde{\mathbf{E}}^r]_{:,j}\|_2} + [\tilde{\mathbf{E}}^r]_{:,j}, \quad (17)$$

where $[\tilde{\mathbf{E}}^r]_{:,j}$ denotes the j -th column of $\tilde{\mathbf{E}}^r$.

Update $\mathcal{G}$: With other variables fixed, the sub-problem for $\mathcal{G}$ is formulated as:

$$\begin{aligned} \min_{\mathcal{G}} \quad & \|\mathcal{G}\|_{\text{TEN}} + \langle \mathcal{Y}, \mathcal{H} - \mathcal{G} \rangle + \frac{\mu}{2} \|\mathcal{H} - \mathcal{G}\|_F^2 \\ \Leftrightarrow \min_{\mathcal{G}} \quad & \frac{1}{\mu} \|\mathcal{G}\|_{\text{TEN}} + \frac{1}{2} \|\mathcal{G} - \tilde{\mathcal{G}}\|_F^2, \end{aligned} \quad (18)$$

where $\tilde{\mathcal{G}} = \mathcal{H} + \mathcal{Y}/\mu$. This Tensor Exponential Norm Minimization problem (TENM) is challenging to solve directly. The optimal solution to Eq. (18) can be obtained through the following theorem:

THEOREM 2. Let $\mathcal{B} \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ with t -SVD $\mathcal{B} = \mathcal{U} * \mathcal{S} * \mathcal{V}^T$ and $\beta > 0$. For the Tensor Exponential Norm minimization problem:

$$\arg \min_{\mathcal{A}} \frac{1}{2} \|\mathcal{A} - \mathcal{B}\|_F^2 + \beta \|\mathcal{A}\|_{\text{TEN}}. \quad (19)$$

Then, optimal solution $\mathcal{A}^*$ is given by

$$\mathcal{A}^* = \mathcal{U} * \text{ifft}(\text{Prox}_{f,\beta}(\mathcal{S}_f), [], 3) * \mathcal{V}^T, \quad (20)$$

where $\text{ifft}(\text{Prox}_{f,\beta}(\mathcal{S}_f), [], 3) \in \mathbb{R}^{n_1 \times n_2 \times n_3}$ is an f -diagonal tensor, with $P_\beta(\mathcal{S}_f^{(k)}(i, i))$ satisfying:

$$\begin{aligned} \text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i)) = \arg \min_{\sigma > 0} \quad & \frac{1}{2} (\sigma - \mathcal{S}_f^{(k)}(i, i))^2 \\ & + \beta \left(1 - e^{-\frac{\mathcal{S}_f^{(k)}(i, i)}{\delta}} \right), \end{aligned} \quad (21)$$

for $1 \leq k \leq n_3$ and $1 \leq i \leq \min(n_1, n_2)$. Notably, $\text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i))$ in $[0, \mathcal{S}_f^{(k)}(i, i)]$, and $\text{Prox}_{f,\beta}(\mathcal{S}_f^{(k)}(i, i)) = 0$ if $\mathcal{S}_f^{(k)}(i, i) = 0$.

The detailed proof of Theorem 2 is provided in the Supplementary. Applying this theorem, we focus on solving the non-convex scalar minimization problem:

$$\min_{x \geq 0} g(x) = \frac{1}{2} (x - y)^2 + \beta f(x), \quad (22)$$

where $f(x) = 1 - e^{-\frac{x}{\delta}}$ with $0 < \delta < 1$. While a closed-form solution for this non-convex problem is elusive, it can be effectively solved using the iterative algorithm proposed in [27]. More importantly, the proposed non-convex penalty function $f(x)$ exactly satisfies

Algorithm 1 Optimization Algorithm of RTGD-MVC

- 1: **Input:** Multi-view data $X^1, X^2, \dots, X^r$, anchor number m , trade-off parameter λ .
- 2: **Output:** $\{H^r\}_{r=1}^v$ for the optimal representation.
- 3: **Initialize:** Set H^r, E^r, S^r to zero matrix, $\mathcal{G} = \mathcal{Y} = 0, \mu = 10^{-5}, \rho = 10^{-4}, \mu_{\max} = \rho_{\max} = 10^{10}, \eta = 2, \epsilon = 10^{-7}$.
- 4: **Repeat**
- 5: Update H^r by Eq.(15).
- 6: Update E^r by Eq.(17).
- 7: Update $\mathcal{G}$ by Eq.(23).
- 8: Update $\mu, \rho, S^r, \mathcal{Y}$ by (24).
- 9: Check the convergence conditions:
 $\|\tilde{F}^r B^r - H^r - E^r\|_{\infty} < \epsilon$ & $\|\mathcal{H} - \mathcal{G}\|_{\infty} < \epsilon$
- 10: **Until** convergence criteria are met.

Assumption 1 of [27]. According to Corollary 1 in [27], the optimal solution is achieved through:

$$\tau^{iter+1} = \left(S_f^k(i, i) - \frac{\partial f(\tau^{iter})}{\mu} \right)_+, \quad (23)$$

where $\tau = \text{Prox}_{f, \beta}(S_f^k(i, i))$, $f(x) = 1 - e^{-\frac{S_f^k(i, i)}{\delta}}$, and $iter$ denotes the iteration number.

Update multipliers and penalty parameter: we update $\{S^r\}_{r=1}^v, \mathcal{Y}, \mu, \rho$ as follows

$$\begin{cases} S^r = S^r + \mu(\tilde{F}^r B^r - H^r - E^r), \\ \mathcal{Y} = \mathcal{Y} + \rho(\mathcal{H} - \mathcal{G}), \\ \mu = \eta\mu, \mu = \min(\mu, \mu_{\max}), \\ \rho = \eta\rho, \rho = \min(\rho, \rho_{\max}), \end{cases} \quad (24)$$

where $\eta = 2 > 1$ accelerates convergence.

Our RTGD-MVC initializes all variables and updates them alternately until convergence. The complete optimization procedure is summarized in Algorithm 1.

Table 2: Dataset Characteristics

Data set	Type	Sam./Clu.	Views
BBC	Text	685 / 5	4
WebKB	Text	1051 / 2	2
Small-Reuters	Text	1500 / 5	6
Reuters	Text	18758 / 6	5
CCV	Video	6773 / 20	3
BDGP	Bioinformatics	2500 / 5	2
Caltech101	Image	1474 / 7	6
Small-NUS	Image	2000 / 31	5
Scene	Image	4485 / 15	3
Nuswideobj	Image	30000 / 31	5
YoutubeFace	Image	101499 / 31	5

4.1 Convergence Analysis

While ADMM algorithms typically address convex problems with two variables, our non-convex, multi-block ADMM's convergence

is theoretically guaranteed by Theorem 3 (proof provided in the supplementary).

THEOREM 3. Let $\{\mathcal{P}_k = (\tilde{B}_k^r, E_k^r, S_k^r, \mathcal{H}_k, \mathcal{G}_k, \mathcal{Y}_k)\}_{k=1}^{\infty}$ be the sequence generated by Algorithm 1. Then:

- $\{\mathcal{P}_k\}_{k=1}^{\infty}$ is bounded.
- Any accumulation point of $\{\mathcal{P}_k\}_{k=1}^{\infty}$ is a KKT point of Eq. (13)

4.2 Complexity analysis

The main time complexity of our RTGD-MVC focuses on four parts: 1) the construction of original bipartite graph, 2) the cross-view diffusion, 3) optimization by iterative solving Eq. (13), and 4) the clustering stage. The first part takes $O(nmd)$, where $d = \sum_{r=1}^v d^r$. The cross-view diffusion part takes $O(m^3v + nm^2v)$. The optimization part is to update the variables (*i.e.* $H^r, E^r, \mathcal{G}$) in problem Eq.(13), which take $O(m^3 + m^2n), O(nd)$, and $O(nmv \log(mv) + nmv^2)$, respectively. Clustering part takes $O(nm^2v^2)$. For $m \ll n$, the total time complexity of RTGD-MVC is $O(nmd + nm^2v + nm^2 + nmv \log(mv))$, which demonstrates linear complexity with respect to the number of samples n . The space complexity is $O((d + mv)n)$, which is linear in n .

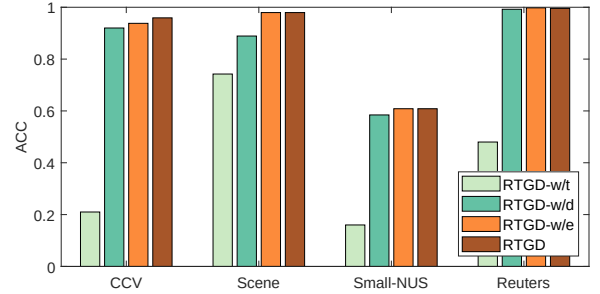


Figure 4: The performance of RTGD-MVC and its three variants: RTGD-w/t, RTGD-w/d, and RTGD-w/e.

5 EXPERIMENT

In this section, we conduct extensive experiments to compare RTGD-MVC with state-of-the-art methods and demonstrate its effectiveness and superiority.

5.1 Experiment setting

5.1.1 Datasets: We evaluate the performance of RTGD-MVC on 11 challenging datasets: BBC, WebKB, Small-Reuters, Reuters, CCV, BDGP, Caltech101, Small-NUS, Scene, Nuswideobj, and YouTubeFace. Details are provided in Table 2.

5.1.2 Baselines: To demonstrate the superiority of our model, we compare RTGD-MVC against nine state-of-the-art multi-view clustering methods, including OPMC [22], FPMVS-CAG [35], FMVACC [33], FDAGF [43], AWMVC [32], FastMICE [9], TLSpNM [8], S2MVTC [26], and RCAGL [24].

5.1.3 Evaluation Metrics: To comprehensively evaluate clustering quality, we employ three widely used metrics: accuracy (ACC), normalized mutual information (NMI), purity (PUR).

5.1.4 Parameter Setting: For the baseline parameters, we adhere to the configurations specified in the corresponding papers and report the best results. For our RTGD-MVC model, the parameter λ and η are searched within $\{10^{-6}, 10^{-5}, \dots, 0\}$, while the number of anchors is explored within $\{2c, 3c, \dots, 8c\}$. Each method is independently run 10 times on each dataset, and we report the average results along with their standard deviations.

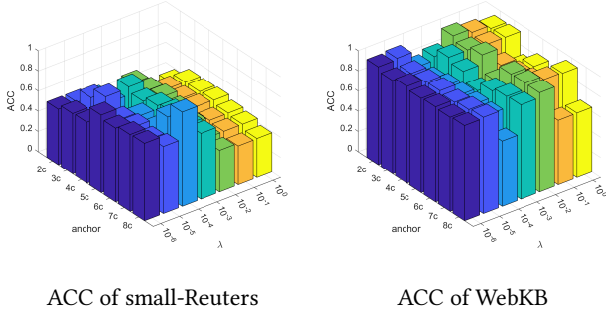


Figure 5: ACC with different parameters λ and anchors m on the WebKB and small-Reuters datasets.

5.2 Experimental Results

The experimental results demonstrate the significant superiority of our proposed RTGD-MVC method in multi-view clustering across diverse datasets. RTGD-MVC consistently achieves the highest performance in terms of accuracy (ACC), normalized mutual information (NMI), and purity (PUR) compared to nine state-of-the-art methods. This success is largely attributed to the innovative integration of graph diffusion, tensor representation, and the Tensor Exponential Norm (TEN) in our framework. Specifically, the graph diffusion mechanism effectively mitigates the propagation and accumulation of graph structure noise, enhancing the robustness of the model. The tensor representation and the TEN further capture high-order correlations within and between views, providing more discriminative and noise-robust representations. For example, on the BDGP dataset, RTGD-MVC attains a perfect ACC of 100%, showcasing its ability to handle complex data structures. These results highlight RTGD-MVC's powerful representational abilities and its effectiveness in uncovering intricate structures in multi-view data.

5.3 Ablation Studies

In this subsection, we compare the results of RTGD-MVC without high-dimensional information at tensor level (RTGD-w/t), RTGD-MVC without high-order information at graph level, derived from cross-view graph diffusion (RTGD-w/d), and RTGD-MVC without the Tensor Exponential Norm, using the standard TNN instead (RTGD-w/e). The results demonstrate that both the high-order structure information at graph level derived from the cross-view graph diffusion model and the high-dimensional structure information at tensor level captured by the improved Tensor Exponential Norm contribute to superior performance.

5.4 Model Analysis

5.4.1 Parameter Analysis: Our RTGD-MVC model has two parameters λ and m , that require tuning. We empirically search for the optimal parameters in each dataset. In Fig.5, we present the ACC results for the Nuswideobj and CCV datasets. It can be observed that when λ is within $\{10^{-6}, \dots, 10^{-3}\}$ and m is in the range of $\{5c, \dots, 8c\}$, the ACC for both NuswideObj and CCV remains consistently high. This indicates that the performance of our RTGD-MVC model is stable across a wide range of parameters, demonstrating its efficiency and robustness.

5.4.2 Convergence Analysis: The convergence of our RTGD-MVC is theoretically guaranteed by the Theorem 3. To further validate its stability, we conducted convergence experiments by recording the values of the stopping criteria. The criteria used are Reconstruction Error (RE), defined as $RE = \max \|\bar{\mathbf{B}}^r - \mathbf{H}^r - \mathbf{E}^r\|_\infty$ and Match Error (ME), defined as $ME = \max \|\mathcal{H} - \mathcal{G}\|_\infty$. We present the results for two datasets in Fig. 7. The values of RE and ME quickly approach 0 within 10 steps and then remain stable, demonstrating the excellent convergence properties of our RTGD-MVC model.

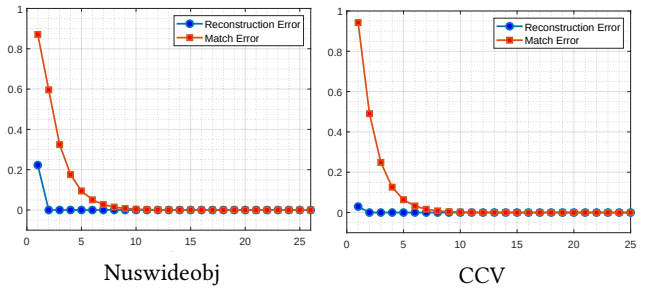


Figure 6: The variation curves of the stopping criteria, namely RE and ME, on two datasets: Nuswideobj and CCV.

6 CONCLUSION

In this paper, we presents a novel and efficient framework termed Robust Tensor Learning with Graph Diffusion for Scalable Multi-view Graph Clustering (RTGD-MVC) to address the challenges of capturing high-order correlations and ensuring noise robustness in multi-view graph clustering. By seamlessly integrating graph diffusion, tensor representation, and the Tensor Exponential Norm (TEN) into a unified optimization model, RTGD-MVC effectively mitigates the propagation and accumulation of graph structure noise. The cross-view diffusion enhances the consistency and complementarity across views, while the improved tensor norm provides robust and discriminative representations. The proposed optimization algorithm with guaranteed convergence and linear complexity further ensures the efficiency and scalability of the model. Extensive experiments on diverse and challenging datasets demonstrate that RTGD-MVC consistently outperforms state-of-the-art methods. RTGD-MVC thus establishes itself as a powerful tool for uncovering intricate structures in multi-view data.

References

- [1] Hongmin Cai, Yu Wang, Fei Qi, Zhuoyao Wang, and Yiu-ming Cheung. 2024. Multiview Tensor Spectral Clustering via Co-regularization. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024).
- [2] Hong Chen, Feiping Nie, Rong Wang, and Xuelong Li. 2023. Fast Unsupervised Feature Selection With Bipartite Graph and L_{2,0}-Norm Constraint. *IEEE Transactions on Knowledge and Data Engineering* 35, 5 (2023), 4781–4793.
- [3] Fan Chung. 2007. The heat kernel as the pagerank of a graph. *Proceedings of the National Academy of Sciences* 104, 50 (2007), 19735–19740.
- [4] Shide Du, Zhiling Cai, Zhihao Wu, Yueyang Pi, and Shiping Wang. 2024. UMCG: Universal Multi-view Consensus Graph Learning with Consistency and Diversity. *IEEE Transactions on Image Processing* (2024).
- [5] Zhibin Gu, Songhe Feng, Zhendong Li, Jiazheng Yuan, and Jun Liu. 2024. NOO-DLE: Joint Cross-View Discrepancy Discovery and High-Order Correlation Detection for Multi-View Subspace Clustering. *ACM Transactions on Knowledge Discovery from Data* 18, 6 (2024), 1–23.
- [6] Zhibin Gu, Zhendong Li, and Songhe Feng. 2024. EDISON: Enhanced Dictionary-Induced Tensorized Incomplete Multi-View Clustering with Gaussian Error Rank Minimization. In *Forty-first International Conference on Machine Learning*.
- [7] Zhibin Gu, Zhendong Li, and Songhe Feng. 2024. Topology-driven multi-view clustering via tensorial refined sigmoid rank minimization. In *Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. 920–931.
- [8] J Guo, Y Sun, J Gao, Y Hu, and B Yin. 2023. Logarithmic Schatten-p Norm Minimization for Tensorial Multi-View Subspace Clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 3 (2023), 3396–3410.
- [9] Dong Huang, Chang-Dong Wang, and Jian-Huang Lai. 2023. Fast multi-view clustering via ensembles: Towards scalability, superiority, and simplicity. *IEEE Transactions on Knowledge and Data Engineering* 35, 11 (2023), 11388–11402.
- [10] Jintian Ji and Songhe Feng. 2023. Anchor structure regularization induced multi-view subspace clustering via enhanced tensor rank minimization. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 19343–19352.
- [11] Jintian Ji and Songhe Feng. 2025. Anchors Crash Tensor: Efficient and Scalable Tensorial Multi-view Subspace Clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2025).
- [12] Zhao Kang, Wangtao Zhou, Zhitong Zhao, Junming Shao, Meng Han, and Zenglin Xu. 2020. Large-scale multi-view subspace clustering in linear time. In *Proceedings of the AAAI conference on artificial intelligence*, Vol. 34. 4412–4419.
- [13] Wei Lan, Tianchuan Yang, Qingfeng Chen, Shichao Zhang, Yi Dong, Huiyu Zhou, and Yi Pan. 2023. Multiview subspace clustering via low-rank symmetric affinity graph. *IEEE Transactions on Neural Networks and Learning Systems* (2023).
- [14] Lusi Li and Haibo He. 2023. Bipartite Graph based Multi-view Clustering. In *2023 IEEE 39th International Conference on Data Engineering (ICDE)*. IEEE, 3889–3890.
- [15] Liang Li, Yuqiang Pan, Jie Liu, Yue Liu, Xinwang Liu, Kenli Li, Ivor W Tsang, and Keqin Li. 2024. Bgae: Auto-encoding multi-view bipartite graph clustering. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [16] Liang Li, Junpu Zhang, Siwei Wang, Xinwang Liu, Kenli Li, and Keqin Li. 2023. Multi-view bipartite graph clustering with coupled noisy feature filter. *IEEE Transactions on Knowledge and Data Engineering* 35, 12 (2023), 12842–12854.
- [17] Xuelong Li, Han Zhang, Rong Wang, and Feiping Nie. 2022. Multiview Clustering: A Scalable and Parameter-Free Bipartite Graph Fusion Method. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 44, 1 (2022), 330–344.
- [18] Yunfan Li, Hancong Li, Yijie Lin, Dan Zhang, Dezhong Peng, Xiting Liu, Jie Xie, Peng Hu, Lu Chen, Han Luo, et al. 2025. MetaQ: fast, scalable and accurate metacell inference via single-cell quantization. *Nature Communications* 16, 1 (2025), 1205.
- [19] Yunfan Li, Dan Zhang, Mouxiang Yang, Dezhong Peng, Jun Yu, Yu Liu, Jiancheng Lv, Lu Chen, and Xi Peng. 2023. scBridge embraces cell heterogeneity in single-cell RNA-seq and ATAC-seq data integration. *Nature communications* 14, 1 (2023), 6045.
- [20] Zhouchen Lin, Minming Chen, and Yi Ma. 2010. The augmented lagrange multiplier method for exact recovery of corrupted low-rank matrices. *arXiv preprint arXiv:1009.5055* (2010).
- [21] Guangcan Liu, Zhouchen Lin, Shuicheng Yan, Ju Sun, Yong Yu, and Yi Ma. 2012. Robust recovery of subspace structures by low-rank representation. *IEEE transactions on pattern analysis and machine intelligence* 35, 1 (2012), 171–184.
- [22] Jiyuan Liu, Xinwang Liu, Xuexiang Yang, Li Liu, Siqi Wang, Weixuan Liang, and Jiangyong Shi. 2021. One-pass multi-view clustering for large-scale data. In *Proceedings of the IEEE/CVF international conference on computer vision*. 12344–12353.
- [23] Suyuan Liu, Ke Liang, Zhibin Dong, Siwei Wang, Xihong Yang, Sihang Zhou, En Zhu, and Xinwang Liu. 2024. Learn from View Correlation: An Anchor Enhancement Strategy for Multi-view Clustering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 26151–26161.
- [24] Suyuan Liu, Qing Liao, Siwei Wang, Xinwang Liu, and En Zhu. 2024. Robust and Consistent Anchor Graph Learning for Multi-View Clustering. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [25] Suyuan Liu, Siwei Wang, Pei Zhang, Kai Xu, Xinwang Liu, Changwang Zhang, and Feng Gao. 2022. Efficient one-pass multi-view subspace clustering with consensus anchors. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 36. 7576–7584.
- [26] Zhen Long, Qiyuan Wang, Yazhou Ren, Yipeng Liu, and Ce Zhu. 2024. S2MVC: a Simple yet Efficient Scalable Multi-View Tensor Clustering. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 26213–26222.
- [27] Canyi Lu, Changbo Zhu, Chunyan Xu, Shuicheng Yan, and Zhouchen Lin. 2015. Generalized singular value thresholding. In *Proceedings of the AAAI conference on artificial intelligence*.
- [28] Huimin Ma, Siwei Wang, Shengju Yu, Suyuan Liu, Jun-Jie Huang, Huijun Wu, Xinwang Liu, and En Zhu. 2024. Automatic and Aligned Anchor Learning Strategy for Multi-View Clustering. In *Proceedings of the 32nd ACM International Conference on Multimedia*. 5045–5054.
- [29] Feiping Nie, Xiaoqian Wang, and Heng Huang. 2014. Clustering and projected clustering with adaptive neighbors. In *Proceedings of the 20th ACM SIGKDD international conference on Knowledge discovery and data mining*. 977–986.
- [30] Yinghui Sun, Xingfeng Li, Sun Quansen, Min-Ling Zhang, and Zhenwen Ren. 2024. Improved Weighted Tensor Schatten p-Norm for Fast Multi-view Graph Clustering. In *ACM Multimedia* 2024.
- [31] Xinhang Wan, Jiyuan Liu, Xinbiao Gan, Xinwang Liu, Siwei Wang, Yi Wen, Tianjiao Wan, and En Zhu. 2024. One-step multi-view clustering with diverse representation. *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [32] Xinhang Wan, Xinwang Liu, Jiyuan Liu, Siwei Wang, Yi Wen, Weixuan Liang, En Zhu, Zhe Liu, and Lu Zhou. 2023. Auto-weighted multi-view clustering for large-scale data. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 10078–10086.
- [33] Siwei Wang, Xinwang Liu, Suyuan Liu, Jiaqi Jin, Wenxuan Tu, Xinzhong Zhu, and En Zhu. 2022. Align then fusion: Generalized large-scale multi-view clustering with anchor matching correspondences. *Advances in Neural Information Processing Systems* 35 (2022), 5882–5895.
- [34] Siwei Wang, Xinwang Liu, Suyuan Liu, Wenxuan Tu, and En Zhu. 2024. Scalable and structural multi-view graph clustering with adaptive anchor fusion. *IEEE Transactions on Image Processing* (2024).
- [35] Siwei Wang, Xinwang Liu, Xinzhong Zhu, Pei Zhang, Yi Zhang, Feng Gao, and En Zhu. 2022. Fast Parameter-Free Multi-View Subspace Clustering With Consensus Anchor Guidance. *IEEE Transactions on Image Processing* 31 (2022), 556–568.
- [36] Wei Xia, Quanxue Gao, Qianqian Wang, Xinbo Gao, Chris Ding, and Dacheng Tao. 2023. Tensorized Bipartite Graph Learning for Multi-View Clustering. *IEEE Transactions on Pattern Analysis & Machine Intelligence* 45, 04 (2023), 5187–5202.
- [37] Yuan Xie, Dacheng Tao, Wensheng Zhang, Yan Liu, Lei Zhang, and Yanyun Qu. 2018. On unifying multi-view self-representations for clustering by tensor multi-rank minimization. *International Journal of Computer Vision* 126 (2018), 1157–1179.
- [38] Ben Yang, Xuetao Zhang, Jinghan Wu, Feiping Nie, Zhiping Lin, Fei Wang, and Badong Chen. 2024. Fast Multiview Anchor-Graph Clustering. *IEEE Transactions on Neural Networks and Learning Systems* (2024).
- [39] Xiaojun Yang, Tuoji Zhu, Danyang Wu, Penglei Wang, Yujia Liu, and Feiping Nie. 2024. Bidirectional Fusion With Cross-View Graph Filter for Multi-View Clustering. *IEEE Transactions on Knowledge and Data Engineering* (2024).
- [40] Shengju Yu, Siwei Wang, Pei Zhang, Miao Wang, Ziming Wang, Zhe Liu, Liming Fang, En Zhu, and Xinwang Liu. 2024. Dvsai: Diverse view-shared anchors based incomplete multi-view clustering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 16568–16577.
- [41] Chao Zhang, Xiuyi Jia, Zechao Li, Chunlin Chen, and Huaxiong Li. 2024. Learning Cluster-Wise Anchors for Multi-View Clustering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 16696–16704.
- [42] Han Zhang, Feiping Nie, and Xuelong Li. 2023. Large-scale clustering with structured optimal bipartite graph. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 8 (2023), 9950–9963.
- [43] Pei Zhang, Siwei Wang, Liang Li, Changwang Zhang, Xinwang Liu, En Zhu, Zhe Liu, Lu Zhou, and Lei Luo. 2023. Let the data choose: Flexible and diverse anchor graph fusion for scalable multi-view clustering. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 37. 11262–11269.
- [44] Runxin Zhang, Shuaijun Hang, Zhenheng Sun, Feiping Nie, Rong Wang, and Xuelong Li. 2024. Anchor-based fast spectral ensemble clustering. *Information Fusion* (2024), 102587.
- [45] Zihua Zhao, Ting Wang, Haonan Xin, Rong Wang, and Feiping Nie. 2024. Multi-view clustering via high-order bipartite graph fusion. *Information Fusion* (2024), 102630.

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009